

RAPA ITI, French Polynesia

WMO: 919580

Lat: 27.618S

Lon: 144.335W

Elev: 7

StdP: 14.69

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	55.7	57.2	44.8	43.8	62.1	46.4	46.7	62.8	27.7	65.1	25.0	66.2	3.0	200	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.3	82.4	75.8	81.3	75.4	80.3	75.0	77.2	80.7	76.4	79.9	75.8	79.3	9.2	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.9	135.5	79.7	75.3	132.5	79.2	74.6	129.4	78.6	40.5	80.8	39.9	80.2	39.2	79.5	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.1	22.5	20.3	DB	52.5	84.1	1.3	1.2	51.6	85.0	50.9	85.7	50.2	86.4	49.3	87.3
			WB	49.9	78.2	1.1	1.0	49.1	79.0	48.4	79.6	47.7	80.1	46.9	80.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.7	74.5	75.9	75.1	72.3	69.5	67.1	65.4	64.7	65.0	66.6	69.3	71.6
	DBStd	4.87	2.64	2.47	2.86	2.86	2.99	3.01	3.00	3.07	3.08	2.75	3.06	3.03
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	147	0	0	0	0	1	13	32	45	38	15	3	0
	CDD50	7197	759	726	778	670	606	512	478	454	450	513	580	671
	CDD65	1872	294	306	313	220	142	76	46	35	38	63	133	206
	CDH74	5837	1240	1739	1571	557	124	11	0	1	4	9	128	453
	CDH80	266	42	106	101	13	0	0	0	0	0	0	0	4
(14) Wind	WSAvg	9.3	9.3	9.4	9.2	10.1	9.0	9.2	9.2	9.2	9.5	8.9	9.0	9.6
(15) Precipitation	PrecAvg	102.8	9.4	7.4	10.2	9.6	9.1	8.6	9.4	9.0	6.9	6.9	6.9	8.9
	PrecMax	148.5	27.4	19.2	27.5	24.4	22.8	16.2	25.1	22.3	23.5	20.5	23.3	29.0
	PrecMin	62.1	1.0	0.5	0.7	1.9	2.0	1.6	3.7	4.4	1.2	0.2	1.2	0.7
	PrecStd	20.0	5.7	4.3	5.5	5.4	5.2	3.4	4.5	3.9	4.3	4.2	4.4	5.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.5	83.8	83.7	81.6	78.1	75.4	73.5	73.5	74.6	75.2	78.5	80.6
		MCWB	75.6	76.3	76.8	75.0	72.4	70.6	68.9	68.8	69.1	70.0	72.4	74.6
	2%	DB	81.2	82.5	82.4	79.7	76.4	73.9	72.5	72.0	72.8	73.8	76.8	79.1
		MCWB	75.2	76.0	75.9	74.2	72.0	69.7	68.2	67.8	68.2	69.0	71.3	74.0
	5%	DB	80.1	81.4	81.1	78.3	75.4	72.9	71.6	71.0	71.5	72.6	75.5	77.9
		MCWB	74.8	75.6	75.6	73.3	71.2	68.7	67.5	66.7	67.3	68.0	70.9	73.4
	10%	DB	78.9	80.2	80.0	77.1	74.4	72.0	70.6	69.9	70.3	71.4	74.2	76.5
		MCWB	74.4	75.0	75.0	72.5	70.2	67.6	66.3	65.4	65.8	66.8	69.9	72.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.2	78.1	78.1	76.8	74.4	72.3	70.8	70.3	70.6	71.7	74.0	76.1
		MCDB	80.4	81.9	82.2	79.9	76.2	74.2	72.4	72.4	73.1	74.1	76.6	78.9
	2%	WB	76.4	77.2	77.2	75.6	73.4	71.0	69.4	68.9	69.3	70.4	72.8	75.1
		MCDB	79.5	80.7	81.0	78.6	75.6	73.2	71.6	71.3	72.0	73.1	75.4	77.9
	5%	WB	75.6	76.4	76.4	74.5	72.3	69.9	68.4	67.7	68.1	69.2	71.8	74.1
		MCDB	78.9	80.0	80.2	77.5	74.8	72.4	71.0	70.4	71.1	72.0	74.7	77.0
	10%	WB	74.9	75.7	75.6	73.4	71.0	68.6	67.2	66.3	66.6	67.8	70.7	73.1
		MCDB	78.3	79.5	79.3	76.5	73.8	71.4	70.1	69.3	69.8	70.6	73.6	76.0
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.3	6.8	6.3	6.7	7.0	6.9	7.2	6.9	7.0	6.6	6.1
		MCDBR	7.9	8.2	8.3	8.1	7.2	7.6	7.2	7.5	7.7	8.1	8.0	7.4
	5% WB	MCWBR	4.3	4.4	4.7	5.2	4.9	5.4	5.6	5.9	5.6	5.6	4.8	4.3
		MCWBR	6.6	7.0	7.4	6.7	5.9	5.8	6.0	6.7	7.0	6.6	6.6	6.1
(40) Clear-Sky Solar Irradiance	taub		0.369	0.372	0.366	0.367	0.355	0.351	0.347	0.347	0.355	0.357	0.355	0.365
		taud	2.623	2.618	2.644	2.627	2.632	2.623	2.623	2.612	2.573	2.566	2.605	2.608
	Ebn at Noon		309	303	295	279	269	262	269	282	293	304	311	311
		Edh at Noon	32	32	29	27	25	24	25	27	31	33	33	33
(44) All-Sky Solar Radiation	RadAvg		2068	1926	1617	1264	988	824	923	1157	1475	1798	2028	1990
	RadStd		226	188	182	87	81	62	86	82	105	108	170	162

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page